

KR0468:

A Clinically Deployable Explainable Multimodal AI System for Oral Cancer Diagnosis and Risk Assessment

Domain: Healthcare, Oncology

Scenario

Existing artificial intelligence approaches for oral cancer detection are predominantly unimodal, relying on either histopathological images or clinical data, and often lack explainability and real-world clinical applicability. This limitation reduces diagnostic reliability and hinders adoption in healthcare settings. Furthermore, current multimodal approaches employ simplistic fusion strategies and fail to integrate heterogeneous data sources such as imaging, clinical, and genomic data in a unified framework. Therefore, there is a need to develop an explainable, scalable, and clinically deployable multimodal deep learning system that integrates histopathological images, intraoral photographs, clinical parameters, and genomic information.

Objective

To design and implement an advanced multimodal architecture using attention-based and transformer-based fusion techniques, combined with robust preprocessing strategies and explainable AI methods. The proposed system will be deployed as a web-based clinical decision support platform, enabling healthcare professionals to upload patient data and obtain accurate diagnosis, risk prediction, and interpretable insights for improved oral cancer management.

Available Datasets Links:

<https://seer.cancer.gov/data/>

<https://portal.gdc.cancer.gov/projects/TCGA-HNSC>

<https://www.ncbi.nlm.nih.gov/geo/>

<https://portal.gdc.cancer.gov/>

<https://www.kaggle.com/datasets/shivam17299/oral-cancer-lips-and-tongue-images>

<https://www.kaggle.com/datasets/zaidpy/oral-cancer-dataset>

<https://data.mendeley.com/datasets/ftmp4cvtmb/1>

Note: You can explore more datasets.